

Questions with Answer Keys

MathonGo

Q1 - 2024 (01 Feb Shift 1)

Let $y = y(x)$ be the solution of the differential equation $\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1, y(0) = 1$.

Then, $\left(\frac{1}{\sqrt{2}} + y\left(\frac{1}{\sqrt{2}}\right)\right)^2$ equals :

(1) $\frac{4}{4+\sqrt{e}}$

(2) $\frac{3}{3-\sqrt{e}}$

(3) $\frac{2}{1+\sqrt{e}}$

(4) $\frac{1}{2-\sqrt{e}}$

Q2 - 2024 (01 Feb Shift 1)

If $x = x(t)$ is the solution of the differential equation $(t+1)dx = (2x + (t+1)^4) dt, x(0) = 2$, then, $x(1)$ equals

Q3 - 2024 (01 Feb Shift 2)

Let α be a non-zero real number. Suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function such that $f(0) = 2$ and $\lim_{x \rightarrow -\infty} f(x) = 1$. If $f'(x) = \alpha f(x) + 3$, for all $x \in \mathbb{R}$, then $f(-\log_e 2)$ is equal to

(1) 3

(2) 5

(3) 9

(4) 7

Q4 - 2024 (01 Feb Shift 2)

If $\frac{dx}{dy} = \frac{1+x-y^2}{y}, x(1) = 1$, then $5x(2)$ is equal to :

Q5 - 2024 (27 Jan Shift 1)

Let $x = x(t)$ and $y = y(t)$ be solutions of the differential equations $\frac{dx}{dt} + ax = 0$ and $\frac{dy}{dt} + by = 0$

respectively, $a, b \in \mathbb{R}$. Given that $x(0) = 2; y(0) = 1$ and $3y(1) = 2x(1)$, the value of t , for which $x(t) = y(t)$, is :

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Questions with Answer Keys

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(1) $\log_{\frac{2}{3}} 2$

(2) $\log_4 3$

(3) $\log_3 4$

(4) $\log_{\frac{4}{3}} 2$

Q6 - 2024 (27 Jan Shift 1)

If the solution of the differential equation $(2x + 3y - 2)dx + (4x + 6y - 7)dy = 0$, $y(0) = 3$, is

$\alpha x + \beta y + 3 \log_e |2x + 3y - \gamma| = 6$, then $\alpha + 2\beta + 3\gamma$ is equal to _____

Q7 - 2024 (27 Jan Shift 2)

If $y = y(x)$ is the solution curve of the differential equation $(x^2 - 4) dy - (y^2 - 3y) dx = 0$,

$x > 2$, $y(4) = \frac{3}{2}$ and the slope of the curve is never zero, then the value of $y(10)$ equals :

(1) $\frac{3}{1+(8)^{1/4}}$

(2) $\frac{3}{1+2\sqrt{2}}$

(3) $\frac{3}{1-2\sqrt{2}}$

(4) $\frac{3}{1-(8)^{1/4}}$

Q8 - 2024 (27 Jan Shift 2)

If the solution curve, of the differential equation $\frac{dy}{dx} = \frac{x+y-2}{x-y}$ passing through the point $(2, 1)$ is

$\tan^{-1}\left(\frac{y-1}{x-1}\right) - \frac{1}{\beta} \log_e \left(\alpha + \left(\frac{y-1}{x-1}\right)^2\right) = \log_e |x-1|$, then $5\beta + \alpha$ is equal to

Q9 - 2024 (29 Jan Shift 1)

A function $y = f(x)$ satisfies

$f(x) \sin 2x + \sin x - (1 + \cos^2 x) f'(x) = 0$ with condition $f(0) = 0$. Then $f\left(\frac{\pi}{2}\right)$ is equal to

(1) 1

(2) 0

(3) -1

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(4) 2

Q10 - 2024 (29 Jan Shift 1)

If the solution curve $y = y(x)$ of the differential equation $(1 + y^2)(1 + \log_e x) dx + x dy = 0$, $x > 0$ passes through the point $(1, 1)$ and $y(e) = \frac{\alpha - \tan\left(\frac{3}{2}\right)}{\beta + \tan\left(\frac{3}{2}\right)}$, then $\alpha + 2\beta$ is

Q11 - 2024 (29 Jan Shift 2)

If $\sin\left(\frac{y}{x}\right) = \log_e |x| + \frac{\alpha}{2}$ is the solution of the differential equation $x \cos\left(\frac{y}{x}\right) \frac{dy}{dx} = y \cos\left(\frac{y}{x}\right) + x$ and $y(1) = \frac{\pi}{3}$, then α^2 is equal to

(1) 3

(2) 12

(3) 4

(4) 9

Q12 - 2024 (30 Jan Shift 1)

Let $y = y(x)$ be the solution of the differential equation $\sec x dy + \{2(1 - x) \tan x + x(2 - x)\} dx = 0$ such that $y(0) = 2$. Then $y(2)$ is equal to :

(1) 2

(2) $2\{1 - \sin(2)\}$ (3) $2\{\sin(2) + 1\}$

(4) 1

Q13 - 2024 (30 Jan Shift 1)

Let $y = y(x)$ be the solution of the differential equation $(1 - x^2) dy = [xy + (x^3 + 2) \sqrt{3(1 - x^2)}] dx$, $-1 < x < 1$, $y(0) = 0$. If $y\left(\frac{1}{2}\right) = \frac{m}{n}$, m and n are coprime numbers, then $m + n$ is equal to _____,

Q14 - 2024 (31 Jan Shift 1)

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The solution curve of the differential equation

$y \frac{dx}{dy} = x (\log_e x - \log_e y + 1)$, $x > 0, y > 0$ passing through the point $(e, 1)$ is

(1) $\left| \log_e \frac{y}{x} \right| = x$

(2) $\left| \log_e \frac{y}{x} \right| = y^2$

(3) $\left| \log_e \frac{x}{y} \right| = y$

(4) $2 \left| \log_e \frac{x}{y} \right| = y + 1$

Q15 - 2024 (31 Jan Shift 1)

Let $y = y(x)$ be the solution of the differential equation

$$\frac{dy}{dx} = \frac{(\tan x) + y}{\sin x (\sec x - \sin x \tan x)},$$

$x \in \left(0, \frac{\pi}{2}\right)$ satisfying the condition $y\left(\frac{\pi}{4}\right) = 2$.

Then, $y\left(\frac{\pi}{3}\right)$ is

(1) $\sqrt{3} (2 + \log_e \sqrt{3})$

(2) $\frac{\sqrt{3}}{2} (2 + \log_e 3)$

(3) $\sqrt{3} (1 + 2 \log_e 3)$

(4) $\sqrt{3} (2 + \log_e 3)$

Q16 - 2024 (31 Jan Shift 2)

The temperature $T(t)$ of a body at time $t = 0$ is $160^\circ F$ and it decreases continuously as per the differential

equation $\frac{dT}{dt} = -K(T - 80)$, where K is positive constant. If $T(15) = 120^\circ F$, then $T(45)$ is equal to

(1) $85^\circ F$

(2) $95^\circ F$

(3) $90^\circ F$

(4) $80^\circ F$

Q17 - 2024 (31 Jan Shift 2)

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Questions with Answer Keys

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Let $y = y(x)$ be the solution of the differential equation

$$\sec^2 x dx + (e^{2y} \tan^2 x + \tan x) dy = 0$$

$$0 < x < \frac{\pi}{2}, y\left(\frac{\pi}{4}\right) = 0. \text{ If } y\left(\frac{\pi}{6}\right) = \alpha,$$

Then $e^{8\alpha}$ is equal to

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Answer Key

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Solutions

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Q1

$$\frac{dy}{dx} = 2x(x+y)^3 - x(x+y) - 1$$

$$x + y = t$$

$$\frac{dt}{dx} - 1 = 2xt^3 - xt - 1$$

$$\frac{dt}{2t^3 - t} = x dx$$

$$\frac{t dt}{2t^4 - t^2} = x dx$$

$$\text{Let } t^2 = z$$

$$\int \frac{dz}{2(2z^2 - z)} = \int x dx$$

$$\int \frac{dz}{4z\left(z - \frac{1}{2}\right)} = \int x dx$$

$$\ln \left| \frac{z - \frac{1}{2}}{z} \right| = x^2 + k$$

$$z = \frac{1}{2 - \sqrt{e}}$$

Q2

$$(t+1)dx = (2x + (t+1)^4) dt$$

$$\frac{dx}{dt} = \frac{2x + (t+1)^4}{t+1}$$

$$\frac{dx}{dt} - \frac{2x}{t+1} = (t+1)^3$$

$$\frac{dx}{dt} - \frac{2x}{t+1} = (t+1)^3$$

$$I \cdot F = e^{-\int \frac{2}{t+1} dt} = e^{-2 \ln(t+1)} = \frac{1}{(t+1)^2}$$

$$\frac{x}{(t+1)^2} = \int \frac{1}{(t+1)^2} (t+1)^3 dt + c$$

$$\frac{x}{(t+1)^2} = \frac{(t+1)^2}{2} + c$$

$$\Rightarrow c = \frac{3}{2}$$

$$x = \frac{(t+1)^4}{2} + \frac{3}{2}(t+1)^2$$

$$\text{put, } t = 1$$

$$x = 2^3 + 6 = 14$$

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Solutions

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Q3

$$f(0) = 2, \lim_{x \rightarrow -\infty} f(x) = 1$$

$$f'(x) - x \cdot f(x) = 3$$

$$\text{I.F} = e^{-\alpha x}$$

$$y(e^{-\alpha x}) = \int 3 \cdot e^{-\alpha x} dx$$

$$f(x) \cdot (e^{-\alpha x}) = \frac{3e^{-\alpha x}}{-\alpha} + c$$

$$x = 0 \Rightarrow 2 = \frac{-3}{\alpha} + c \Rightarrow \frac{3}{\alpha} = c - 2 \dots (1)$$

$$f(x) = \frac{-3}{\alpha} + c \cdot e^{\alpha x}$$

$$x \rightarrow -\infty \Rightarrow 1 = \frac{-3}{\alpha} + c(0)$$

$$\alpha = -3 \therefore c = 1$$

$$f(-\ln 2) = \frac{-3}{\alpha} + c \cdot e^{\alpha x}$$

$$= 1 + e^{3 \ln 2} = 9$$

(But α should be greater than 0 for finite value of c)

Q4

$$\frac{dx}{dy} - \frac{x}{y} = \frac{1-y^2}{y}$$

$$\text{Integrating factor} = e^{\int -\frac{1}{y} dy} = \frac{1}{y}$$

$$x \cdot \frac{1}{y} = \int \frac{1-y^2}{y^2} dy$$

$$\frac{x}{y} = \frac{-1}{y} - y + c$$

$$x = -1 - y^2 + cy$$

$$x(1) = 1$$

$$1 = -1 - 1 + c \Rightarrow c = 3$$

$$x = -1 - y^2 + 3y$$

$$5x(2) = 5(-1 - 4 + 6)$$

$$= 5$$

Q5

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Solutions

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$$\frac{dx}{dt} + ax = 0$$

$$\frac{dx}{x} = -adt$$

$$\int \frac{dx}{x} = -a \int dt$$

$$\ln|x| = -at + c$$

$$\text{at } t = 0, x = 2$$

$$\ln 2 = 0 + c$$

$$\ln x = -at + \ln 2$$

$$\frac{x}{2} = e^{-at}$$

$$x = 2e^{-at} \dots (i)$$

$$\frac{dy}{dt} + by = 0$$

$$\frac{dy}{y} = -bdt$$

$$\ln|y| = -bt + \lambda$$

$$t = 0, y = 1$$

$$0 = 0 + \lambda$$

$$y = e^{-bt} \dots (iv)$$

According to question

$$3y(1) = 2x(1)$$

$$3e^{-b} = 2(2e^{-a})$$

$$e^{a-b} = \frac{4}{3}$$

$$\text{For } x(t) = y(t)$$

$$\Rightarrow 2e^{-at} = e^{-bt}$$

$$2 = e^{(a-b)t}$$

$$2 = \left(\frac{4}{3}\right)^t$$

$$\log_{\frac{4}{3}} 2 = t$$

Q6

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Solutions

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$$2x + 3y - 2 = t \quad 4x + 6y - 4 = 2t$$

$$2 + 3 \frac{dy}{dx} = \frac{dt}{dx} \quad 4x + 6y - 7 = 2t - 3$$

$$\frac{dy}{dx} = \frac{(2x + 3y - 2)}{4x + 6y - 7}$$

$$\frac{dt}{dx} = \frac{-3t + 4t - 6}{2t - 3} = \frac{t - 6}{2t - 3}$$

$$\int \frac{2t - 3}{t - 6} dt = \int dx$$

$$\int \left(\frac{2t - 12}{t - 6} + \frac{9}{t - 6} \right) \cdot dt = x$$

$$2t + 9 \ln(t - 6) = x + c$$

$$2(2x + 3y - 2) + 9 \ln(2x + 3y - 8) = x + c$$

$$x = 0, y = 3$$

$$c = 14$$

$$4x + 6y - 4 + 9 \ln(2x + 3y - 8) = x + 14$$

$$x + 2y + 3 \ln(2x + 3y - 8) = 6$$

$$\alpha = 1, \beta = 2, \gamma = 8$$

$$\alpha + 2\beta + 3\gamma = 1 + 4 + 24 = 29$$

Q7

$$(x^2 - 4) dy - (y^2 - 3y) dx = 0$$

$$\Rightarrow \int \frac{dy}{y^2 - 3y} = \int \frac{dx}{x^2 - 4}$$

$$\Rightarrow \frac{1}{3} \int \frac{y - (y - 3)}{y(y - 3)} dy = \int \frac{dx}{x^2 - 4}$$

$$\Rightarrow \frac{1}{3} (\ln |y - 3| - \ln |y|) = \frac{1}{4} \ln \left| \frac{x - 2}{x + 2} \right| + C$$

$$\Rightarrow \frac{1}{3} \ln \left| \frac{y - 3}{y} \right| = \frac{1}{4} \ln \left| \frac{x - 2}{x + 2} \right| + C$$

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Solutions

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$$\text{At } x = 4, y = \frac{3}{2}$$

$$\therefore C = \frac{1}{4} \ln 3$$

$$\therefore \frac{1}{3} \ln \left| \frac{y-3}{y} \right| = \frac{1}{4} \ln \left| \frac{x-2}{x+2} \right| + \frac{1}{4} \ln(3)$$

$$\text{At } x = 10$$

$$\frac{1}{3} \ln \left| \frac{y-3}{y} \right| = \frac{1}{4} \ln \left| \frac{2}{3} \right| + \frac{1}{4} \ln(3)$$

$$\ln \left| \frac{y-3}{y} \right| = \ln 2^{3/4}, \forall x > 2, \frac{dy}{dx} < 0$$

$$\text{as } y(4) = \frac{3}{2} \Rightarrow y \in (0, 3)$$

$$-y + 3 = 8^{1/4} \cdot y$$

$$y = \frac{3}{1 + 8^{1/4}}$$

Q8

$$\frac{dy}{dx} = \frac{x+y-2}{x-y}$$

$$x = X + h, y = Y + k$$

$$\frac{dY}{dX} = \frac{X+Y}{X-Y}$$

$$\left. \begin{array}{l} h+k-2=0 \\ h-k=0 \end{array} \right\} h=k=1$$

$$Y = vX$$

$$v + \frac{dv}{dX} = \frac{1+v}{1-v} \Rightarrow X - \frac{dv}{dX} = \frac{1+v^2}{1-v}$$

$$\frac{1-v}{1+v^2} dv = \frac{dX}{X}$$

$$\tan^{-1} v - \frac{1}{2} \ln(1+v^2) = \ln |X| + C$$

As curve is passing through (2, 1)

$$\tan^{-1} \left(\frac{y-1}{x-1} \right) - \frac{1}{2} \ln \left(1 + \left(\frac{y-1}{x-1} \right)^2 \right) = \ln |x-1|$$

$$\therefore \alpha = 1 \text{ and } \beta = 2$$

$$\Rightarrow 5\beta + \alpha = 11$$

Q9

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Solutions

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$$\frac{dy}{dx} - \left(\frac{\sin 2x}{1 + \cos^2 x} \right) y = \sin x$$

$$\text{I.F.} = 1 + \cos^2 x$$

$$y \cdot (1 + \cos^2 x) = \int (\sin x) dx$$

$$= -\cos x + C$$

$$x = 0, C = 1$$

$$y\left(\frac{\pi}{2}\right) = 1$$

Q10

$$\int \left(\frac{1}{x} + \frac{\ln x}{x} \right) dx + \int \frac{dy}{1 + y^2} = 0$$

$$\ln x + \frac{(\ln x)^2}{2} + \tan^{-1} y = C$$

$$\text{Put } x = y = 1$$

$$\therefore C = \frac{\pi}{4}$$

$$\Rightarrow \ln x + \frac{(\ln x)^2}{2} + \tan^{-1} y = \frac{\pi}{4}$$

$$\text{Put } x = e$$

$$\Rightarrow y = \tan\left(\frac{\pi}{4} - \frac{3}{2}\right) = \frac{1 - \tan \frac{3}{2}}{1 + \tan \frac{3}{2}}$$

$$\therefore \alpha = 1, \beta = 1$$

$$\Rightarrow \alpha + 2\beta = 3$$

Q11

Differential equation :-

$$x \cos \frac{y}{x} \frac{dy}{dx} = y \cos \frac{y}{x} + x$$

$$\cos \frac{y}{x} \left[x \frac{dy}{dx} - y \right] = x$$

Divide both sides by x^2

$$\cos \frac{y}{x} \left(\frac{x \frac{dy}{dx} - y}{x^2} \right) = \frac{1}{x}$$

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Solutions

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$$\text{Let } \frac{y}{x} = t$$

$$\cos t \left(\frac{dt}{dx} \right) = \frac{1}{x}$$

$$\cos t dt = \frac{1}{x} dx$$

Integrating both sides

$$\sin t = \ln |x| + c$$

$$\sin \frac{y}{x} = \ln |x| + c$$

$$\text{Using } y(1) = \frac{\pi}{3}, \text{ we get } c = \frac{\sqrt{3}}{2}$$

$$\text{So, } \alpha = \sqrt{3} \Rightarrow \alpha^2 = 3$$

Q12

$$\frac{dy}{dx} = 2(x-1) \sin x + (x^2 - 2x) \cos x$$

Now both side integrate

$$y(x) = \int 2(x-1) \sin x dx + \left[(x^2 - 2x) (\sin x) - \int (2x - 2) \sin x dx \right]$$

$$y(x) = (x^2 - 2x) \sin x + \lambda$$

$$y(0) = 0 + \lambda \Rightarrow 2 = \lambda$$

$$y(x) = (x^2 - 2x) \sin x + 2$$

$$y(2) = 2$$

Q13

$$\frac{dy}{dx} - \frac{xy}{1-x^2} = \frac{(x^3+2)\sqrt{3(1-x^2)}}{1-x^2}$$

$$\text{IF} = e^{-\int \frac{x}{1-x^2} dx} = e^{+\frac{1}{2} \ln(1-x^2)} = \sqrt{1-x^2}$$

$$y\sqrt{1-x^2} = \sqrt{3} \int (x^3 + 2) dx$$

$$y\sqrt{1-x^2} = \sqrt{3} \left(\frac{x^4}{4} + 2x \right) + c$$

$$\Rightarrow y(0) = 0 \therefore c = 0$$

$$y\left(\frac{1}{2}\right) = \frac{65}{32} = \frac{m}{n}$$

$$m + n = 97$$

Q14

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Solutions

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$$\frac{dx}{dy} = \frac{x}{y} \left(\ln\left(\frac{x}{y}\right) + 1 \right)$$

$$\text{Let } \frac{x}{y} = t \Rightarrow x = ty$$

$$\frac{dx}{dy} = t + y \frac{dt}{dy}$$

$$t + y \frac{dt}{dy} = t(\ln(t) + 1)$$

$$y \frac{dt}{dy} - t \ln(t) \Rightarrow \frac{dt}{t \ln(t)} = \frac{dy}{y}$$

$$\Rightarrow \int \frac{dt}{t \cdot \ln(t)} = \int \frac{dy}{y}$$

$$\Rightarrow \int \frac{d}{p} = \int \frac{dy}{y} \quad \text{let } \ln t = p$$

$$\frac{1}{t} dt = dp$$

$$\Rightarrow \ln p = \ln y + c$$

$$\ln(\ln t) = \ln y + c$$

$$\ln\left(\ln\left(\frac{x}{y}\right)\right) = \ln y + c$$

$$\text{at } x = e, y = 1$$

$$\ln\left(\ln\left(\frac{e}{1}\right)\right) = \ln(1) + c \Rightarrow c = 0$$

$$\ln\left|\ln\left(\frac{x}{y}\right)\right| = \ln y$$

$$\ln\left(\frac{x}{y}\right) = e^{ky}$$

$$\left|\ln\left(\frac{x}{y}\right)\right| = y$$

Q15

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Solutions

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$$\begin{aligned}\frac{dy}{dx} &= \frac{\sin x + y \cos x}{\sin x \cdot \cos x \left(\frac{1}{\cos x} - \sin x \cdot \frac{\sin x}{\cos x} \right)} \\ &= \frac{\sin x + y \cos x}{\sin x (1 - \sin^2 x)} \\ \frac{dy}{dx} &= \sec^2 x + y \cdot 2(\operatorname{cosec} 2x) \\ \frac{dy}{dx} - 2 \operatorname{cosec}(2x) \cdot y &= \sec^2 x \\ \frac{dy}{dx} + p \cdot y &= Q\end{aligned}$$

$$\text{IF.} = e^{\int x dx} = e^{\int -2 \cos x (2x) dx}$$

Let $2x = t$

$$2 \frac{dx}{dt} = 1$$

$$dx = \frac{dt}{2}$$

$$= e^{-\int \cos \cos(t) a}$$

$$= e^{-|\tan \frac{t}{2}|}$$

$$= e^{-t|\sin x} = \frac{1}{|\tan x|}$$

$$y(\text{IF}) = \int Q(\text{IF}) dx + c$$

$$\Rightarrow y \frac{1}{|\tan x|} = \int \sec^2 x \cdot \frac{1}{|\tan x|} + c$$

$$y \cdot \frac{1}{|\tan x|} = \int \frac{dt}{|t|} + c \quad \text{for } \tan x = t$$

$$y \cdot \frac{1}{|\tan x|} = \ln |t| + c$$

$$y = |\tan x|(\ln |\tan x| + c)$$

$$\text{Put } x = \frac{\pi}{4}, y = 2$$

$$2 = \ln 1 + c \Rightarrow c = 2$$

$$y = |\tan x|(\ln |\tan x| + 2)$$

$$y\left(\frac{\pi}{3}\right) = \sqrt{3}(\ln \sqrt{3} + 2)$$

Q16

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Solutions

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$$\begin{aligned}\frac{dT}{dt} &= -k(T - 80) \\ \int_{160}^T \frac{dT}{(T - 80)} &= \int_0^t -K dt \\ [\ln |T - 80|]_{160}^T &= -kt \\ \ln |T - 80| - \ln 80 &= -kt \\ \ln \left| \frac{T - 80}{80} \right| &= -kt \\ T &= 80 + 80e^{-kt} \\ 120 &= 80 + 80e^{-k \cdot 15} \\ \frac{40}{80} &= e^{-k \cdot 15} = \frac{1}{2} \\ \therefore T(45) &= 80 + 80e^{-k \cdot 45} \\ &= 80 + 80(e^{-k \cdot 15})^3 \\ &= 80 + 80 \times \frac{1}{8} \\ &= 90\end{aligned}$$

Q17

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Solutions

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$$\sec^2 x \frac{dx}{dy} + e^{2y} \tan^2 x + \tan x = 0$$

$$\left(\text{Put } \tan x = t \Rightarrow \sec^2 x \frac{dx}{dy} = \frac{dt}{dy} \right)$$

$$\frac{dt}{dy} + e^{2y} \times t^2 + t = 0$$

$$\frac{dt}{dy} + t = -t^2 \cdot e^{2y}$$

$$\frac{1}{t^2} \frac{dt}{dy} + \frac{1}{t} = -e^{2y}$$

$$\left(\text{Put } \frac{1}{t} = u \Rightarrow \frac{-1}{t^2} \frac{dt}{dy} = \frac{du}{dy} \right)$$

$$\frac{-du}{dy} + u = -e^{2y}$$

$$\frac{du}{dy} - u = e^{2y}$$

$$\text{I.F.} = e^{-\int dy} = e^{-y}$$

$$ue^{-y} = \int e^{-y} \times e^{2y} dy$$

$$\frac{1}{\tan x} \times e^{-y} = e^y + c$$

$$x = \frac{\pi}{4}, y = 0, c = 0$$

$$x = \frac{\pi}{6}, y = \alpha$$

$$\sqrt{3}e^{-\alpha} = e^{\alpha} + 0$$

$$e^{2\alpha} = \sqrt{3}$$

$$e^{8\alpha} = 9$$

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